# Improving Urban Cycling Safety and Comfort through Optimized Infrastructure Upgrades

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***Abstract*— This paper proposes a bi-level optimization framework to improve urban cycling infrastructure by upgrading bike lanes' degree of separation from motorized traffic. The six-level degree of separation classification reflects increasing safety and comfort as cyclists' exposure to annual average daily traffic decreases. The lower level estimates cyclist flows using an all-or-nothing assignment method that accounts for factors like slope, angular changes, and traffic, ensuring realistic flow distribution. The upper level minimizes traffic exposure by upgrading segments to higher DoS levels within a fixed budget. A case study on Grenoble's cycling network applies this framework using a modified genetic algorithm to optimize safety, comfort, and budget efficiency. Results are compared to a specialized algorithm designed to improve overall perceived safety and comfort. Findings highlight that targeted infrastructure upgrades enhance both local and network-wide safety, better balancing cyclist flows. This framework provides urban planners with a data-driven tool for prioritizing cycling infrastructure investments.**

***Index Terms*— Cycling infrastructure, bi-level optimization, degree of separation, annual average daily traffic, safety, comfort, urban planning.**

## I. Introduction

Cycling provides significant health, environmental, and economic benefits, including improved cardiovascular health, lower emissions, and reduced congestion [1], [2], [3]. To encourage cycling, authorities promote it as an alternative to short-distance car travel. However, perceived safety and comfort strongly influence ridership [4]. Studies show that physically separated lanes enhance safety perception, boosting cycling rates [5], [6]. Well-connected, high-quality infrastructure further increases cycling adoption [7], [8]. Given budget constraints, cities must strategically invest in infrastructure to maximize safety and comfort where it has the greatest impact.

*Related Work:* The dynamics of bicycle flow and optimization of cycling infrastructure have received considerable attention in recent literature. In [9], [10], bi-level optimization approaches are used to maximize cycling distances while minimizing car travel times, balancing benefits for both cyclists and motorists. Additionally, [11] evaluates gradual network expansions through societal cost-benefit analyses.

Graph-theory optimization techniques have also been applied to enhance cycling network connectivity that aims to improve cyclist experience. For example, [12] presents a framework that minimizes upgrade costs by prioritizing connectivity, ensuring that all origin-destination pairs for cyclists are accessible via routes that meet a specified level of service. Similarly, [13] proposes an optimization model that balances shorter routes with budget constraints. In [14], an all-or-nothing bicycle traffic assignment approach is applied to model cyclist flows, considering factors like slope and angular changes.

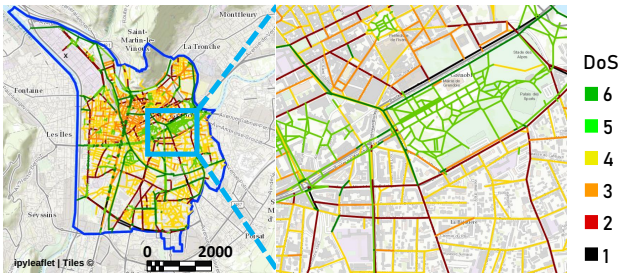



Fig. 1: Cycling network of Grenoble, France. The degree of separation (DoS) score of the cycling infrastructure from motorized vehicles is indicated in ascending order from 1 to 6.

While existing methods provide useful insights, they generally do not prioritize cyclist comfort and safety through targeted infrastructure upgrades. Our framework fills this gap with a bi-level approach: the upper level strategically upgrades bike lanes to reduce cyclists' exposure to traffic, taking into account infrastructure costs and lane characteristics, while the lower level evaluates how these upgrades affect cyclist flow distributions. This approach captures the interdependent effects of safety, comfort, and flow dynamics—an integrated framework that, to the best of our knowledge, has not been previously developed.

*Statement of Contributions:* This paper proposes a modeling and bi-level optimization framework for upgrading cycling infrastructure based on a six-category classification of bike lanes by degree of separation (DoS) from motor traffic. Higher DoS levels correspond to greater safety and comfort, reducing exposure to car traffic. Our framework focuses on upgrading lanes from lower to higher DoS categories to maximize cyclist comfort and safety. Key contributions include: (1) estimating cyclist flows with an all-or-nothing assignment method that incorporates factors like slope, angular changes, and annual average daily traffic (AADT); (2) formulating a bi-level optimization problem from a utilitarian perspective to minimize cyclists' exposure

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| DoS | Description | Geovelo | AADT (veh/day) | Mean speed (km/hr) |
|---|---|---|---|---|
| 6 | Dedicated bike lanes, fully separated from motor traffic. | 7, 14 | 14 400 | 50 |
| 5 | Bike lanes shared with pedestrians, separate from cars. | 2, 10, 11, 15, 16, 17, 20 | 16 000 | 50 |
| 4 | Mixed-traffic roads (30 km/h) without designated bike lanes. | 9, 12, 13, 21, 22 | 960 | 30 |
| 3 | Shared roads with cycling pictograms. | 5, 8 | 3 120 | 30 |
| 2 | Roads with painted bike lanes. | 1, 3, 4, 6 | 6 912 | 40 |
| 1 | Unmarked shared car-bike lanes. | 18, 19 | 8 448 | 40 |

TABLE I: The first column represents the degree of separation (DoS), which categorizes bike lane infrastructure based on safety levels. The second column describes the category of cycling infrastructure. The third column lists the corresponding Geovelo classification numbers. The fourth and fifth columns present data from the GTLVille platform in Grenoble, detailing annual average daily traffic (AADT) and car mean speed in the adjacent roads to cycling paths, respectively.

to traffic, while accounting for the redistribution of flows due to infrastructure upgrades, as applied to the Grenoble cycling network; and (3) to solve the problem, we modify a standard genetic algorithm aiming at budget efficiency to achieve a balanced solution that respects budget constraints. We then compare its results with those of an algorithm specifically designed to improve overall perceived safety and comfort by maximizing the largest connected component (CC) [15].

*Organization:* The remainder of this paper is organized as follows: Section II describes the classification of bike lanes and the graph representation of the network. Section III details the model design, including the flow assignment method, the bi-level optimization problem, the algorithms used to address it. Section IV presents a case study of the Grenoble cycling network. Section V concludes the paper and outlines directions for future research.

## II. Network Data and Features

This section outlines the construction of the cycling network using a six-category DoS classification. These scores serve as link weights in the network graph and define the hierarchy for modeling upgrades to safer and more comfortable categories.

### A. Bike Lane Classification

Based on Geovelo's platform, which classifies bike lanes using OpenStreetMap (OSM) data, the DoS categories are condensed into six levels to represent safety and comfort variations according to physical separation from traffic [15]. The methodology is openly available online[1]. Table I outlines the DoS categories and their correspondence with Geovelo's classification, where numbers are assigned alphabetically. High-DoS lanes (DoS 6 and 5) consistently achieve the highest safety and comfort scores due to full separation from motor traffic, while lower-DoS lanes become less safe and comfortable as AADT increases. Fig. 1 illustrates these classifications in Grenoble, France.

### B. Graph Representation

Let the network be represented by a weighted directed graph (digraph) $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, where $\mathcal{V}$ is the set of nodes corresponding to intersections or road endpoints, $\mathcal{A}$ is the set of directed arcs representing roads, and $\mathcal{W}$ is the set of weights assigned to those arcs. For each arc $a = (u, v) \in \mathcal{A}$, with $u, v \in \mathcal{V}$, the weight $w_a \in \mathcal{W}$ represents the DoS associated with the link. The weights are assumed to be symmetric, meaning $w_{(u,v)} = w_{(v,u)}$. The geographical length of each arc $a$ is denoted by $d_a$.

In [15], the authors demonstrated that the DoS classification effectively represents safety and comfort perceptions. Upgrading arcs to higher DoS levels locally enhances safety by reducing traffic exposure, encouraging cyclists to shift toward safer routes. This, in turn, leads to improved overall safety and comfort across the network.

## III. Model Design

In this section, we introduce the model for predicting cyclist flows, followed by the bi-level optimization problem, which aims to minimize cyclists' exposure to traffic while accounting for the distribution of cycling flows across the network to improve safety and comfort. Finally, we then present the two algorithms employed to solve this NP-hard problem.

### A. Cyclist Flows

| Symbol | Description | Domain | Unit |
|---|---|---|---|
| $q_{(u,v)}$ | Bike demand from $u$ to $v$ | $\mathbb{R}_{\geq 0}$ | bike/day |
| $p_{(u,v)}$ | Shortest path between $u$ and $v$ | $\mathcal{A}$ | - |
| $d_a$ | Arc $a$ length/distance | $\mathbb{R}_{>0}$ | m |
| $\measuredangle_a$ | Angular change at $a$ | $\mathbb{R}_{\geq 0}$ | deg |
| $s_a$ | Slope factor of arc $a$ | $\mathbb{R}_{\geq 0}$ | - |
| $f_a$ | Traffic factor on arc $a$ | $\mathbb{R}_{\geq 0}$ | - |
| $b_{(u,v)}$ | Cyclist distance $u$-$v$ | $\mathbb{R}_{\geq 0}$ | m |
| $x_a$ | Flow on arc $a$ | $\mathbb{R}_{\geq 0}$ | bike/day |
| $x_p$ | Flow on path $p$ | $\mathbb{R}_{\geq 0}$ | bike/day |
| $\sigma_{p_{(u,v)}}(a)$ | arc $a$ is part of shortest path $p$? | $\{0, 1\}$ | - |
| $\rho, \kappa, \psi$ | Calibration parameters | $\mathbb{R}$ | - |
| $x_a^{\text{AADT}}$ | AADT adjacent to $a$ | $\mathbb{R}_{\geq 0}$ | veh/day |
| $w_a$ | Original DoS of $a$ | $\mathbb{R}_{\geq 0}$ | - |
| $\tilde{w}_a$ | Candidate DoS for $a$ | $\mathbb{R}_{\geq 0}$ | - |
| $\pi(w_a, \tilde{w}_a)$ | Upgrade cost for $a$ | $\mathbb{R}_{\geq 0}$ | € |
| $J_{\mathcal{E}}$ | Total cyclist exposure | $\mathbb{R}_{\geq 0}$ | - |

TABLE II: Abbreviated notation for variables in cycling network model.

Following [9], we use an all-or-nothing (AON) traffic assignment model under the assumption of no congestion, where all demand $q_{(u,v)}$ from the EMD2020 survey [16] is assigned to the shortest available path between nodes $u, v \in \mathcal{V}$. This approach, demonstrated in [9] and implemented in [14], simplifies flow assignment by prioritizing the shortest path. Additionally, OpenStreetMap and cycling

[1] https://github.com/leunamCampero/ITSC2024

flow data from Cardiff, UK, are used to calibrate parameters, define a custom distance metric incorporating cycling infrastructure, and apply betweenness centrality to estimate flow distribution. The method is formally presented below.

To incorporate cycling-specific factors,a modified distance metric is defined. Let $u, v \in \mathcal{V}$ be nodes, and let $p_{(u,v)} = (a_1, \ldots, a_k)$ represent the shortest path between $u$ and $v$. For an arc $a \in p_{(u,v)}$, let $d_a$ denote the geographical length; $\measuredangle_a$ represents the angular change (in degrees) along $a$ if it is the first or last arc in the path, or at junctions between consecutive arcs; $s_a$ the slope factor based on the terrain slope $m_a$; and $f_a$ the traffic factor, representing car traffic conditions on roads adjacent to bike lanes, which can significantly impact bike flow on lanes that are shared with car traffic.

The cyclist distance $b_{(u,v)}$ between nodes $u$ and $v$ is then defined as

$$b_{(u,v)} = \sum_{a \in p_{(u,v)}} (d_a \cdot s_a \cdot (f_a)^{\rho}) + \sum_{a \in p_{(u,v)}} \measuredangle_a \cdot \kappa. \quad (1)$$

The traffic factor $f_a$ is defined as

$$f_a = \begin{cases} \psi \cdot \exp\left(x_a^{\text{AADT}}\right) & \text{if } w_a < 5, \\ \psi \cdot 2.03 & w_a = 5, \\ \psi & w_a = 6, \end{cases} \quad (2)$$

where $x_a^{\text{AADT}}$ and $w_a$ represent the annual average daily traffic of the car lanes adjacent to bike arc $a$ expressed in 1000s, and the DoS for bike arc $a$, respectively. According to [17], the effect of AADT is neutralized if the cycling lane is fully separated from car lanes (i.e., when DoS $= 5$ or 6). The factor of 2.03 signifies more than twice the probability of bike crash risk in spaces shared with pedestrians compared to fully separated infrastructure, as identified in [18].

Let $p_{(u,v)}$ be the shortest path between nodes $u$ and $v$ based on the cyclist distance in (1). Since the origin-destination (OD) demand in our data is symmetric, each node acts as both an origin and a destination, ensuring balanced inflow and outflow. The flow conservation for paths is given by

$$x_{p_{(u,v)}} = q_{(u,v)}, \quad (3)$$

where $q_{(u,v)}$ represents the demand between nodes $u$ and $v$. Next, let $\sigma_{p_{(u,v)}}(a)$ be a binary variable indicating whether arc $a$ is part of the shortest path $p_{(u,v)}$. Then, the total cyclist flow on arc $a$, denoted as $x_a$, is

$$x_a = \sum_{u,v \in \mathcal{V}} q_{(u,v)} \cdot \sigma_{p_{(u,v)}}(a) \geq 0, \quad \forall a \in \mathcal{A}. \quad (4)$$

This ensures that the total flow on each arc is the sum of all demands for trips passing through it, maintaining balance across the network due to the symmetry of the demand matrix.

To assess cyclist exposure to traffic conditions, we introduce $\mathcal{E}$ as a general measure of exposure, defined for pahts in the network. This metric helps evaluate network improvements. For a shortest path $p_{(u,v)} = (a_1, \ldots, a_k)$, cyclist exposure captures the combined effect of traffic-related risk and the distance of cyclist interaction with traffic, and is defined as

$$\mathcal{E}_{p_{(u,v)}} = \sum_{a \in p_{(u,v)}} d_a \cdot f_a, \quad (5)$$

where $d_a$ and $f_a$ are the normalized arc length and traffic factor, respectively. For a single arc $a \in \mathcal{A}$, cyclist exposure simplifies to

$$\mathcal{E}_a = d_a \cdot f_a \in [0, 1]. \quad (6)$$

Next, we introduce the bi-level optimization frameworks for improving cyclist safety. The lower-level problem estimates cyclist flows via AON assignment, incorporating slope and traffic factors. The upper-level optimization selects lane upgrades to reduce cyclist exposure while incorporating both utilitarian and sufficientarian transport justice principles. Additionally, we introduce the algorithms designed to solve these problems efficiently.

### *B. Optimization Problem*

Taking a utilitarian perspective, we aim to reduce cyclists' average exposure to vehicular traffic while accounting for their flow distribution across the network. By prioritizing upgrades that enhance the DoS within a fixed budget $B$, we seek to maximize the average safety and comfort of cyclists, directing flows toward safer routes and reducing high-traffic exposure for all network users.

Each upgrade to the cycling infrastructure, such as increasing a road segment's DoS, incurs a cost per meter. DoS upgrades are discrete and must satisfy a fixed budget $B$. We define the following notations:

- $\mathbf{w}_0 = \left[w_1, \ldots, w_a, \ldots, w_{|\mathcal{A}|}\right]^\top \in \mathbb{N}^{|\mathcal{A}|}$: the original arc weights (DoS) of the graph $\mathcal{G}$;
- $\tilde{\mathbf{w}} = \left[\tilde{w}_1, \ldots, \tilde{w}_a, \ldots, \tilde{w}_{|\mathcal{A}|}\right]^\top \in \Omega$: the candidate weights to be optimized, whereby

$$\Omega = \left\{ \tilde{w} \in \mathbb{N}^{|\mathcal{A}|} \colon w_a \leq \tilde{w}_a \leq 6, \forall a \in \{1, \ldots, |\mathcal{A}|\} \right\}; \quad (7)$$

Additionally, we define the following, where the values are normalized to be scaled between their respective minimum and maximum values:

- $\mathbf{b}(\tilde{\mathbf{w}}) = \left[b_1, \cdots, b_a, \cdots, b_{|\mathcal{A}|}\right]^\top \in \mathbb{R}_{\geq 0}^{|\mathcal{A}|}$: the vector of normalized cyclist distances, defined in (1);
- $\mathbf{x}^\star(\tilde{\mathbf{w}}) = \left[x_1, \cdots, x_a, \cdots, x_{|\mathcal{A}|}\right]^\top \in \mathbb{R}_{\geq 0}^{|\mathcal{A}|}$: the vector of normalized arc flows resulting from the AON assignment, computed according to (3) and (4);
- $\boldsymbol{\mathcal{E}}(\tilde{\mathbf{w}}) = \left[\mathcal{E}_1, \cdots, \mathcal{E}_a, \cdots, \mathcal{E}_{|\mathcal{A}|}\right]^\top \in \mathbb{R}_{\geq 0}^{|\mathcal{A}|}$: the vector of traffic exposure, which is also a function of the chosen upgrades $\tilde{\mathbf{w}}$, as defined in (6);
- $\pi(w_a, \tilde{w}_a) \in \mathbb{R}_{\geq 0}$: the cost per meter of upgrading arc $a$ from its original DoS $w_a$ to the new DoS $\tilde{w}_a$;
- $J_{\mathcal{E}} = \boldsymbol{\mathcal{E}}^\top(\tilde{\mathbf{w}})\mathbf{x}^\star(\tilde{\mathbf{w}}) \in \mathbb{R}_{\geq 0}$: the total cyclist exposure throughout the network, where both $\boldsymbol{\mathcal{E}}$ and $\mathbf{x}^\star$ depend on the chosen DoS levels $\tilde{\mathbf{w}}$.

The optimization problem has two interconnected levels:

*Upper-Level Problem* ($\mathbb{U}$): This problem represents the strategic infrastructure planning goal. Here, we aim to minimize cyclist exposure to car traffic by choosing optimal upgrades to the DoS across the network, while staying within the budget $B$. The objective function minimizes the total exposure defined as the product of arc length, traffic factor, and cyclist flow. This ensures that upgrades prioritize long arcs with high traffic factors and flows, as they present the greatest risk of exposure to cyclists.

*Lower-Level Problem* ($\mathbb{L}$): This problem represents the flow distribution, which models how cyclists will travel across the network once the upgrades have been made. The objective is to determine the optimal distribution of cyclist flows across the network by assigning each origin-destination (OD) pair to the path that minimizes cyclist distance by an all-or-nothing assignment as in Section III-A. This distribution depends on the cyclist-specific distance metric, which reflects the comfort and safety of each route.

Because changes in the DoS at the upper level influence the cyclist distance (and thus the flows) at the lower level, the two levels are nested. This means that every decision made at the upper level affects the solution of the lower level.

*Problem 1: (Bi-level network-upgrade optimization problem):*

$$\mathbb{U}: \quad \mathbf{w}^\star = \arg\min_{\tilde{\mathbf{w}}\in\Omega} \; J_{\mathcal{E}}(\tilde{\mathbf{w}}) = \boldsymbol{\mathcal{E}}^\top(\tilde{\mathbf{w}})\mathbf{x}^\star(\tilde{\mathbf{w}}) \tag{8}$$

$$\text{s.t.} \sum_{a=1}^{|\mathcal{A}|} d_a \cdot \pi(w_a, \tilde{w}_a) \leq B \tag{9}$$

$$\mathbb{L}: \quad \mathbf{x}^\star(\tilde{\mathbf{w}}) = \arg\min_{\mathbf{x}\in\mathbb{R}^{|\mathcal{A}|}_{\geq 0}} \; \mathbf{b}(\tilde{\mathbf{w}})^\top \mathbf{x} \tag{10}$$

$$\text{s.t. } x_a = \sum q \cdot \sigma(a) \geq 0 \quad \forall a \in \{1, \ldots, |\mathcal{A}|\} \tag{11}$$

as in (4).

The bi-level optimization Problem 1 is NP-hard because it involves selecting a subset of upgrades for the arcs while respecting a fixed budget, similar to the 0-1 knapsack problem. The challenge lies in balancing the upgrades' costs and their impact on safety and comfort, which requires combinatorial selection and may be computationally intractable for large instances.

### C. Solution Algorithm

To address the computational challenges stemming from Problem 1, we leverage/devise a genetic algorithm and a maximize largest CC algorithm.

*1) Genetic algorithm with efficient budget spending (GAB):* This algorithm optimizes cyclist traffic exposure while adhering to budget constraints. It first generates a set of possible upgrade solutions, evaluates their effectiveness, and refines them by combining solutions and applying random mutations. Configurations exceeding the budget are discarded, while under-budget solutions are expanded by upgrading lower-DoS links until funds are fully utilized. Though this heuristic approach does not guarantee a global optimum, it efficiently finds a near-optimal solution within a reasonable time. Algorithm 1 outlines the full procedure with the following notation:

- $\tilde{B}$: Remaining budget after the initial solution $\tilde{\mathbf{w}}$ has been evaluated.
- $\Delta_a$: The upgrade cost for arc $a$, computed as $\pi(\tilde{w}_a, w_{\text{DoS}_6})$, which is the cost to increase the DoS to its maximum value.

**Algorithm 1** GAB

**Input:** $\tilde{B}$, $\tilde{\mathbf{w}}$
**Output:** Updated arc weights: $\tilde{w}_a$, remaining budget: $\tilde{B}$
**for** $a \in \mathcal{A}$ sorted by DoS$(a)$ **do**
  Compute upgrade cost: $\Delta_a = \pi(\tilde{w}_a, \tilde{w}_{\text{DoS}_6})$
  **if** $\tilde{B} \geq \Delta_a$ **then**
    Upgrade arc $\tilde{w}_a$ to $w_{\text{DoS}} = 6$.
    Update remaining budget: $\tilde{B} \leftarrow \tilde{B} - \Delta_a$
  **end if**
  **if** $\tilde{B} \leq 0$ **then**
    **Exit**
  **end if**
**end for**
**Return:** Updated $\tilde{w}_a$, remaining $\tilde{B}$

*2) Maximize largest connected component algorithm (MLCC):* This algorithm optimizes the cycling network by upgrading edges to enhance connectivity within budget limits [15]. It prioritizes upgrading the largest connected component (CC) with the highest DoS to DoS=6. If multiple CCs share this highest DoS, they are connected in order of size using the shortest DoS-path, where arcs are weighted by $b_a \cdot (7 - w_a)$ to prioritize higher-DoS links. Once a primary high-DoS network is established, other CCs are iteratively linked and upgraded until the budget is exhausted or all edges reach DoS=6. While the algorithm does not explicitly account for cyclist flows, it indirectly improves route choices by favoring high-DoS paths. With polynomial complexity, it efficiently enhances perceived safety and comfort in large cycling networks. Algorithm 2 outlines the full procedure with the following notation:

- $\mathcal{C}$: Largest connected sub-network with DoS $= 6$.
- $\mathcal{C}_\gamma$: Connected component (CC) with DoS level $\gamma \leq 6$.

**Algorithm 2** MLCC Algorithm

**Input:** Budget $B$, graph $\mathcal{G}$.
**Output:** Upgraded DoS levels for arcs $\mathcal{A}$
**while** $B > 0$ and $\exists$ connected component $\mathcal{C}_\gamma$ with DoS $\gamma \leq 6$ **do**
  Find shortest DoS-path between $\mathcal{C}$ and $\mathcal{C}_\gamma$
  Upgrade edges along path to DoS $= 6$ and update $B$
  Upgrade edges in $C_\gamma$ to DoS $= 6$ and extend $\mathcal{C}$
**end while**
**Return:** Final DoS levels for all edges in $\mathcal{A}$.

## IV. Grenoble Cycling Network Case Study

In this section, we present the case study of Grenoble, France, and discuss the optimization results achieved by the GAB and MLCC algorithms. We then apply the largest CC metric to compute the overall improvements in perceived safety and comfort resulting from both algorithms.

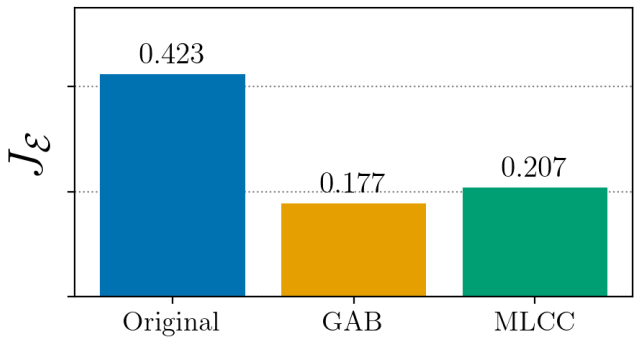



Fig. 2: Initial traffic exposure and reduced exposure after optimization using the GAB and MLCC algorithms for Problem 1 with a budget constraint of 50%, i.e., $\beta = \frac{1}{2}$.

### A. Parameters and Algorithm Comparisons

To solve the all-or-nothing traffic assignment problem and determine cyclist flows, as described in Section III-A, we use parameters that are better suited for the model, as suggested by [14]. Specifically, we set $\rho = 0.04$, $\kappa = 0.14$ and $\psi = 0.84$. The slope factor $s_a$ is determined in Table III, where downhill routes are considered as the reciprocal of the slope factor when traversing in the opposite direction.

| Slope ($m_a$) % | Uphill Factor ($s_a$) | Downhill Factor ($s_a$) |
|---|---|---|
| $m_a < 2$ | 1.0 | 1.0 |
| $2 \leq m_a < 4$ | 1.371 | $1/1.371 \approx 0.729$ |
| $4 \leq m_a < 6$ | 2.203 | $1/2.203 \approx 0.454$ |
| $m_a \geq 6$ | 4.239 | $1/4.239 \approx 0.236$ |

TABLE III: Slope factors for cycling distance calculation

Let $B = \beta \cdot B_{\text{max}}$, be the available budget for optimization, with $B_{\text{max}}$ represents the budget required to upgrade the entire network to DoS $= 6$, and $\beta \in [0, 1]$. The cost per meter to upgrade the infrastructure along an edge is set to reflect different construction costs between levels, following Cerema's cost [19] studies. For this analysis, we assume $\beta = \frac{1}{2}$, representing 50% of the maximum budget.

To solve Problem 1, with Algorithm 1 we adapt the standard genetic algorithm from Pygmo [20]. Fig. 2 compares the original cyclist exposure to traffic of $0.42$ with the optimized results of $0.17$ and $0.20$ from the GAB and MLCC approaches, respectively. The results demonstrate that GAB is more effective at minimizing traffic exposure compared to MLCC, even though GAB only guarantees a local optimum. As shown in Table IV, while GAB results in 141 km of bike lanes with DoS=6 and MLCC achieves 170 km, GAB allocates nearly 10 times more bike lanes with DoS=5. This allocation strategy optimizes budget usage by prioritizing a category that is less costly than the highest one while still significantly reducing cyclist exposure to traffic.

Although MLCC does not explicitly consider flow, it still indirectly lowers traffic exposure due to the all-or-nothing traffic assignment model, which assumes cyclists follow the shortest path based on the cyclist distance metric (1). This metric heavily depends on the DoS of each lane, and MLCC specializes in connecting origin-destination pairs with high DoS lanes, providing safe and comfortable paths.

| DoS (Km) | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| Original | 31.0 | 43.9 | 146.9 | 36.3 | 43.8 | 5.9 |
| GAB | 141.8 | 74.0 | 59.0 | 19.9 | 13.2 | 0 |
| MLCC | 170.5 | 6.1 | 105.3 | 16.3 | 7.9 | 1.2 |

TABLE IV: Number of kilometers of bike lanes for each DoS

Fig. 3 shows the DoS levels and cyclist flow distribution before and after optimization. Both algorithms effectively upgrade the network, improving connectivity and promoting safer, more efficient routes. Future work will refine optimization, consider cyclist congestion, and develop algorithms closer to the global optimum.

## V. Conclusions

This work presented a framework for upgrading cycling infrastructure to enhance safety and comfort within budget constraints. Using a genetic algorithm to efficiently allocate resources and an algorithm to improve connectivity among the safest routes, we show that targeted upgrades reduce cyclists' exposure to traffic and improve flow distribution.

Future work will focus on taking into account factors such as potential congestion and lane capacity, as well as the impact on other road users, possibly in intermodal settings [21]. In addition to upgrading infrastructure, we aim to integrate justice considerations into the optimization process to ensure the equitable distribution of safety and comfort enhancements across the network [22].

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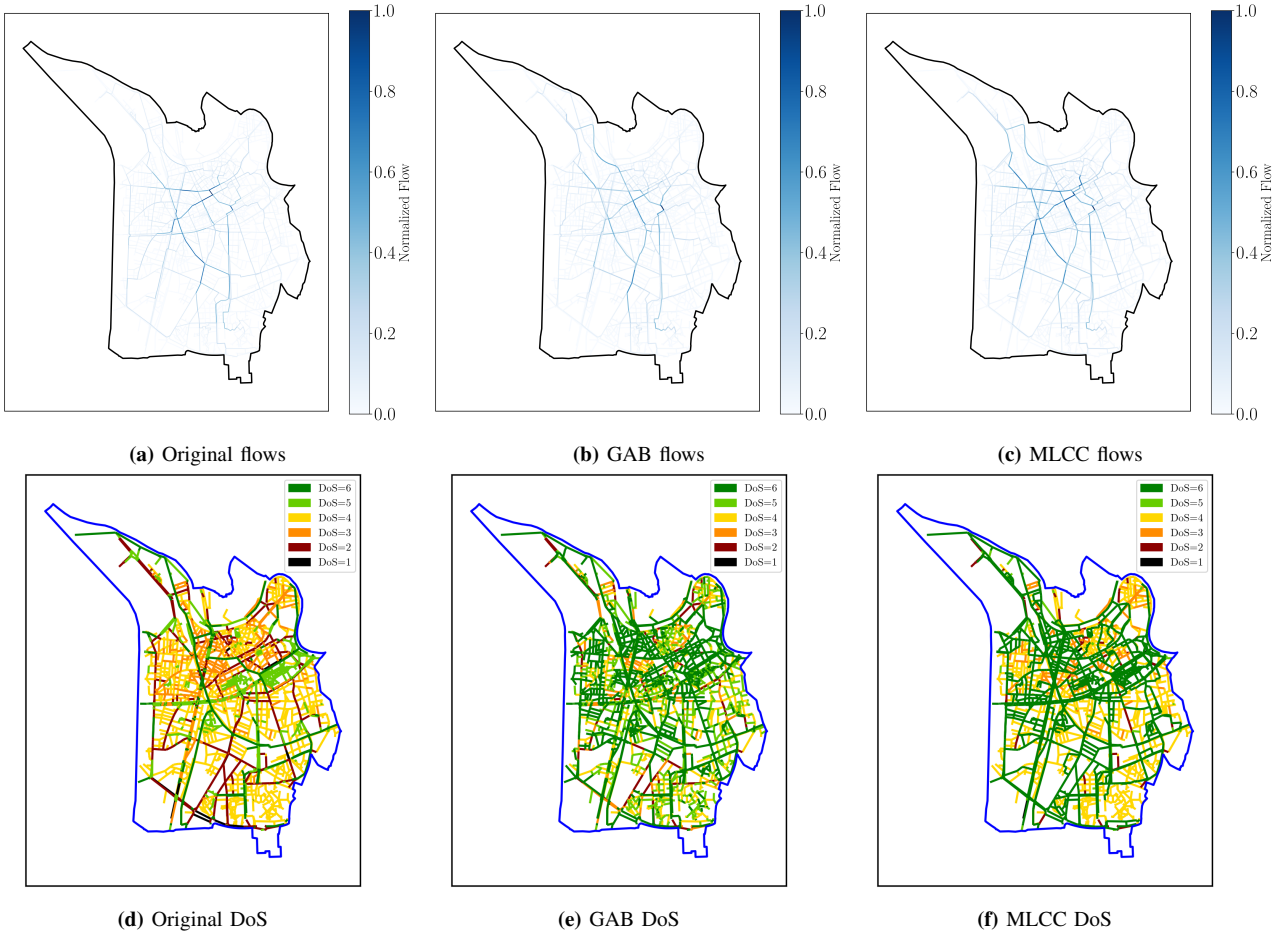
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**(a)** Original flows **(b)** GAB flows **(c)** MLCC flows

**(d)** Original DoS **(e)** GAB DoS **(f)** MLCC DoS

Fig. 3: The upper section shows the flow mapping, while the lower section shows the DoS configuration. The optimization algorithms operate with 50% of the total budget needed for a complete network upgrade.


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